

BUS 123 - Solving Business Problems with Technology

MATH-M01-L01

BUS 123 — MATH-M01— Whole Numbers, Fractions & Percents

PRE-READING

BUS 123 — MATH-M01— Whole Numbers, Fractions & Percents

BUS 123 · MATH-M01-L01 · WHOLE NUMBERS, FRACTIONS & PERCENTS

Course: Solving Business Problems with Technology · Fall 2026 **Track:** MATH · **Module:** M01 ·

1 · CONNECT TO PRIOR KNOWLEDGE

This module is the mathematical foundation for everything that follows in BUS 123. Every formula you will build — markup percentages, payroll calculations, breakeven analysis, and loan math — starts with the skills you'll practice here: whole number operations, fraction-to-decimal conversion, and percent calculations.

This should feel very familiar; for this class pay close attention to the **business context** and the **Excel approach**, which add layers that go beyond a standard algebra class.

Our case study company for this module is **Meridian Advisory Group**, a wealth management firm whose day-to-day operations generate exactly the kinds of arithmetic problems we will solve.

2 · CORE CONCEPTS

Part A — Whole Number Operations

The four arithmetic operations — addition, subtraction, multiplication, and division — are the building blocks of business math. In a business context, each operation maps to a specific type of question:

Operation	Business Meaning	Excel Formula
Addition	Summing revenue across accounts	<code>=SUM(B2:B10)</code> or <code>=B2+B3</code>
Subtraction	Net income = Revenue – Expenses	<code>=B5 – C5</code>
Multiplication	Total fees = Accounts × Fee per account	<code>=B3 * C3</code>
Division	Per-client fee = Total ÷ Client count	<code>=B7 / C7</code>

Worked Example — Meridian Advisory Group:

Meridian begins Q3 with 48 client accounts, adds 12 new clients, and loses 3. The management fee is \$4,200 per account per quarter.

- Quarter-end accounts: $48 + 12 - 3 = 57$
- Total quarterly fees: $57 \times \$4,200 = \$239,400$
- In Excel: `=B2+B3-B4` for the account count, then `=D5*C2` for total fees

Always reference cells rather than typing numbers directly into the formula.

⚠ Unit Mismatch Warning

Before multiplying, check that both numbers use the same unit and time period. Multiplying a quarterly account count by an annual fee rate gives a meaningless result. Label every column header with its unit (e.g., `Fee/Account/Quarter`) to catch this error early.

Part B — Fractions in Context

A fraction expresses a part of a whole: the **numerator** is the piece being measured, and the **denominator** is the total universe. In business, fractions appear as staff allocations, market share figures, and service-line proportions.

Excel converts fractions to decimals automatically when you divide the numerator by the denominator; format the result cell as a percentage or fraction if needed.

Worked Example — Meridian Advisory Group:

Meridian's 6-person advisory team is split evenly: 2 advisors each in Tax, Estate Planning, and Investment Management.

- Each service line represents $2/6 = 1/3 \approx 0.3333$ of total capacity.
- If Meridian hires 3 more Tax advisors, the Tax team grows to 5 of 9 total advisors: $5/9 \approx 0.5556 = 55.6\%$.
- Note that the denominator changed from 6 to 9 — **you must use the new total in the denominator**.

✅ Tip: The Denominator Is the Whole Universe

When the total changes (e.g., new hires), update the denominator before recalculating. Using an outdated denominator is one of the most common fraction errors in business analysis.

Part C — Percent Calculations

A percent is a fraction expressed out of 100. Business math involves **three distinct percent calculations**, each requiring a different formula structure. Confusing them is a frequent source of errors:

Operation	Excel Formula	What It Calculates
Case 1 — % of a Whole	<code>=whole * rate</code>	What is 15% of \$239,400? → <code>=239400*0.15</code>
Case 2 — % Change	<code>=(new - old) / old</code>	Fees: \$1.2M → \$1.255M → <code>=(1255000-1200000)/1200000</code>
Case 3 — Ratio as %	<code>=part / whole</code>	4 of 25 clients renewed → <code>=4/25</code> (format as %)

Worked Example — Forward and Backward:

Meridian earned \$1,200,000 in AUM fees last year and projects a 7.5% increase.

- Projected fees: `=1200000*(1+0.075)` = **\$1,290,000**
- Actual fees came in at \$1,255,000
- Actual percent change: `=(1255000-1200000)/1200000` = **4.58%** — below the 7.5% target

In Excel, always use cell references: $= (B5 - B4) / B4$. Format the result cell as **Percentage** — do not manually multiply by 100.

⚠ Common Mistake: % Change vs. % of Total

These are different formulas.

- **% change** compares a value to its earlier self: $(\text{New} - \text{Old}) / \text{Old}$

- **% of total** compares a part to a whole at one point in time: $\text{Part} / \text{Whole}$

Ask yourself: am I measuring how something **changed over time**, or what **share** it represents of a bigger number?

3 · FORMULA REFERENCE

Operation	Excel Formula	What It Calculates
SUM range	<code>=SUM(B2:B10)</code>	Add all values in a range
Arithmetic ops	<code>=B2+B3-B4</code>	Add/subtract individual cells
Multiply	<code>=B3 * C3</code>	Qty × Price, Accounts × Fee
Divide	<code>=B7 / C7</code>	Total ÷ Count = per-unit
Fraction → Decimal	<code>=B2 / C2</code>	Numerator ÷ Denominator
% of Whole	<code>=whole * rate</code>	e.g., <code>=B4 * 0.15</code>
% Change	<code>=(new - old) / old</code>	Base = older number always
Ratio as %	<code>=part / whole</code>	Format result as Percentage
Grow by rate	<code>=B4 * (1 + rate)</code>	e.g., <code>=B4 * (1 + 0.075)</code>

4 · CHECK YOUR UNDERSTANDING

Answer each question before looking at the answer key below. Show your Excel formula approach, not just the arithmetic result.

1. Meridian starts a quarter with 57 accounts and loses 4. What is the new account total? Write the Excel formula.
1. 2. Total quarterly fee revenue was \$239,400. Tax services generated \$85,000 of that. What fraction — and what percent — of total fees came from Tax?
1. 3. Meridian is targeting a 12% increase in total AUM fees from a base of \$239,400/quarter. What is the quarterly fee target? Write the Excel formula.
1. 4. Actual fees came in at \$268,200. The prior quarter was \$239,400. What was the percent change? Which formula case does this represent?
1. 5. Meridian's Investment team of 2 advisors represents what fraction of a 9-person total team? Express as a decimal rounded to 4 places.

1. 6. True or False: the denominator in a percent change calculation should always be the newer (larger) number.
1. 7. Meridian runs 4 service lines. Two lines each have 15 clients, one has 18 clients, one has 9 clients. What percent of total clients does the line with 18 clients represent?

Answer Key · Check Your Understanding

#	Answer
1	53 accounts. Excel: <code>=57-4</code> or <code>=B2-B3</code> . Unit check: both are counts of accounts.
2	Fraction: $85,000/239,400 \approx 0.3551$. Percent: $\approx 35.51\%$. Excel: <code>=85000/239400</code> → format as Percentage. This is Case 3 (ratio as %).
3	\$268,128. Excel: <code>=239400*(1+0.12)</code> or <code>=B4*(1+0.12)</code> . This is Case 1 (% of whole, grow-by variant).
4	Percent change = $(268,200 - 239,400) / 239,400 \approx 12.03\%$. Excel: <code>=(268200-239400)/239400</code> . This is Case 2 (% change). The base is the older number (\$239,400).
5	$2/9 \approx 0.2222$. Excel: <code>=2/9</code> . Note: the denominator is 9 (total team), not 6 (old team).
6	False. The denominator in a percent change calculation should always be the older (earlier) number.
7	18 clients out of $(15+15+18+9) = 57$ total. $18/57 \approx 31.58\%$. Excel: <code>=18/57</code> → format as Percentage. This is Case 3.

5 · KEY VOCABULARY

Term	Definition
Numerator	The top number in a fraction; represents the part being measured.
Denominator	The bottom number in a fraction; represents the total or whole.
Percent	A ratio expressed as a fraction out of 100. Written with the % symbol.
Percent Change	A measure of how much a value has grown or declined relative to its starting point. Formula: <code>(New - Old) / Old</code> .
Percent of Total	A ratio comparing a part to the whole at a single point in time. Formula: <code>Part / Whole</code> .
Cell Reference	In Excel, using a cell address (e.g., <code>B4</code>) in a formula rather than typing a number directly, so the formula recalculates automatically when input changes.
Unit Mismatch	An error that occurs when two values being multiplied or compared use different units or time periods (e.g., mixing annual and quarterly rates).
AUM Fees	Assets Under Management fees — revenue a financial advisory firm earns based on the total value of client assets it manages.